CS 598 LLMs Post Pre-training

Project Final Presentation

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ILLINOIS

Project Topic

Personalizing Stylistic Text Generation with Parameter-Efficient Fine Tuning and Retrieval Augmented Generation

Motivation

Importance of Personalized Text Generation:

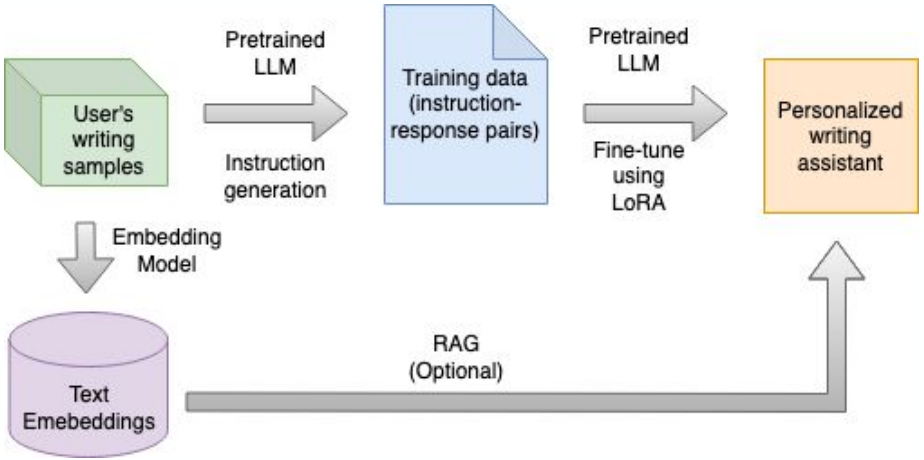
- Enhances user satisfaction by generating content that mirrors their unique style.
- Reduces the time spent editing or rewriting generated content for stylistic consistency.
- Personalization adapts to diverse linguistic styles and goals, improving its relevance and flexibility in different contexts than traditional LLMs are capable of.
- Personalized LLMs have many applications in social media, customer service, creative writing, marketing, etc.

Challenges in Personalization:

- High computational cost associated with user specific model fine-tuning.
- Difficulty in balancing the incorporation of user specific styles with maintaining coherence and fluency.
- Limited capability to dynamically include external context relevant to the task.

Proposed Idea

- Given a user's writing samples, use a pre-trained LLM to generate instruction-response pairs for fine-tuning.
- Apply LoRA based parameter-efficient fine-tuning on the pre-trained LLM using the instruction-response pairs such that the model learns the user's writing style.
- During text generation, optionally employ retrieval-augmented generation to provide additional context to the model.
- We apply to this to varied domains such as social media posts, user reviews and creative writing.



Background

Parameter Efficient Fine-tuning (PEFT)

- Fine-tuning to adapt LLMs with fewer trainable parameters
- Methods include LoRA and RoSA
- Optimizes memory usage, allowing for personalization even on low-resource devices
- Model accuracy is maintained

Retrieval Augmented Generation (RAG)

- Combines a pre-trained language model with a retrieval system for relevant context
- Enhances personalization by retrieving similar documents during model training and inference
- Improves content relevance by grounding generated outputs in user-related context
- Useful when model needs to leverage past data to generate accurate, context-aware responses

Lewis, Patrick et al. "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks." ArXivabs/2005.11401 (2020)
Hu, J. Edward et al. "LoRA: Low-Rank Adaptation of Large Language Models." ArXivabs/2106.09685 (2021)

Methodology Overview

3 main steps:

1. **Data generation:** Generate instruction-response pairs from user-provided writing samples.
2. **Fine-tuning:** Perform LoRA-based fine-tuning on instruction-response pairs to personalize the LLM.
3. **Generation phase:** Retrieve contextually relevant documents using RAG during text generation.

We chose 2 users each from 5 different domains, and fine-tuned a total of 10 models using both Llama and Mistral.

Dataset Creation

User Data Collection:

- Collected writing samples from the user.
- Preprocessed the data through cleaning (removal of special characters, URLs, etc.) and deduplication.

Instruction-Completion Pair Generation:

- Utilized a pre-trained LLM to process the writing samples and generate instruction-completion pairs.
- Prompts for instruction generation were meticulously designed to capture stylistic nuances.
- These serve as inputs for fine-tuning the model to replicate the user's writing style.
- Example:
    - Reddit comment: "Just burned 1200 calories. Forgot the pizza in the oven."
    - Generated Instruction: "Write a concise, humorous comment about pizza getting burnt in the oven."

Tokenization:

- Examine text length distributions to set appropriate truncation and padding thresholds.
- Tokenized both instructions and completions using the pre-trained model's tokenizer.
- Token-level preprocessing to address discrepancies like out-of-vocabulary tokens or excessive padding.

Instruction Generation Prompts

Blog Posts:
“Summarize the following blog post by capturing its main theme, key events, or personal reflections. Frame the summary as an instruction to write a similar blog post and start with 'Instruction:'. Ensure the summary retains the tone and focus of the post, whether it’s casual, reflective, or humorous, without copying the content verbatim. Here is the blog post: ”

Reddit Comments:
“Summarize the following Reddit comment by inferring the likely sentiment, topic, or intent based on its content alone. Use an imperative form and aim for one concise sentence that captures the essence of the comment. Ensure the summary looks like an instruction to write a similar comment and starts with "Instruction:". Here is the Reddit comment: “

Amazon Reviews:
“Summarize the following product review in an imperative form, in one or two or maximum three sentences, ensuring the summary captures key sentiments, features, and feedback (positive or negative) about the product. Avoid copying the review verbatim but retain the essence. The summary should look like an instruction for writing a similar review and start with 'Instruction:'. Here is the product review: “

Instruction Generation Prompts

IMDb Reviews:
“Summarize the following movie review by identifying its core sentiment, key points about the film (such as plot, acting, direction, or visuals), and any standout observations. Frame the summary as an instruction to write a similar review. Ensure the summary reflects the review’s tone—whether enthusiastic, critical, or balanced—without copying the content verbatim. Here is the review:”

Book Excerpts:
“Summarize the following book excerpt by capturing its central theme, notable events or ideas, and any unique stylistic elements present in the text. Frame the summary as an instruction to write a similar excerpt. Ensure the summary preserves the tone—whether descriptive, suspenseful, or poetic—without copying the content verbatim. Here is the excerpt:”

Model Fine-tuning

Model setup

- Loaded the base model (Mistral or Llama)
- Applied LoRA (Low-Rank Adaptation) for parameter-efficient fine-tuning

Quantization

- Enabled 4-bit quantization to optimize memory usage
  - Model parameters were loaded in 4-bit precision
  - Allowed training with limited GPU compute

LoRA configuration

- Targeted attention layers: q_proj and v_proj
- Integrated LoRA settings with a custom dropout to improve generalization:
  - Rank (r): 16
  - Alpha: 32
  - Dropout: 10%

Training

- **User-Specific Fine-Tuning:**
  - Personalized models were trained for 10 users across 5 distinct domains.
  - The training configurations were specific for each of these models.
- **Training Configuration:**
  - Used appropriate epoch settings and batch sizes optimized based on each dataset's size.
  - Number of epochs varied between 150 - 400
  - Used small batch sizes (usually 4) to accommodate the small size of user datasets.
  - Used small learning rates (e.g., in the range of 1e-4 to 1e-5) for stability during fine-tuning.
  - Used 8-bit AdamW optimizer for faster training
  - Conducted systematic hyperparameter tuning of various other parameters to balance training efficiency and model generalization.
- **Overfitting Mitigation:**
  - Closely monitored validation loss during training to prevent overfitting, particularly critical given the small dataset sizes.
  - Early stopping techniques were employed to halt training once diminishing returns in validation performance were observed.

RAG Inference

Context embedding with SentenceTransformer

- Embedded user inputs using the all-MiniLM-L6-v2 SentenceTransformer model.

Context retrieval with FAISS

- Built a FAISS index for fast similarity search over the embedded user inputs.
- Retrieved the top 3 most relevant matches for a given query.

Prompt engineering with retrieved context

- Integrated user inputs to create a context-aware prompt for the LM.
- Ensured the prompt included additional context but was still coherent and intent-preserving.

Response generation using the fine-tuned model

- Used the fine-tuned model to generate responses that aligned with the query and the retrieved context.
- Used top-k sampling to improve response diversity and relevance.

Datasets Used

PushShift Reddit Comments

- 1.7 billion Reddit comments
- We used 600 comments each for two users

Blog Authorship Corpus

- 681.3K blog posts written by more than 19,000 authors on blogger.com in 2004
- We used 800 posts each for two users

Amazon Reviews

- Contains over 200 million product reviews on Amazon published between 1996 and 2018.
- We used 600 reviews each for two users

IMDb62

- 62,000 movie reviews written by 62 prolific users of IMDb
- We used 1000 reviews each for two users

BookMIA

- 9,870 rows of books and snippets from the books
- We used 100 excerpts each from the books David Copperfield and Little Women

Models Used

Pre-trained Models:

1. Mistral-7B-Instruct
2. Meta-Llama-3-8B

These models were chosen because they are open-source and offer high performance despite their smaller size, making them computationally efficient for fine-tuning with limited resources. This is ideal for training a personalized writing assistant.

Retrieval Embedding Model: all-MiniLM-L6-v2

Metrics Used

BLEU Score:

- It measures the n-gram overlap between generated text and reference text
- It can assess whether the generated text uses similar phrases or word sequences as the reference, like distinctive collocations or idiomatic expressions

ROUGE Score:

- It measures recall for overlapping sequences (ROUGE-1, ROUGE-2, ROUGE-L)
- It can indicate how well the generated text covers the key elements of the reference style, such as rhetorical devices, lexical diversity, or certain repeated motifs.

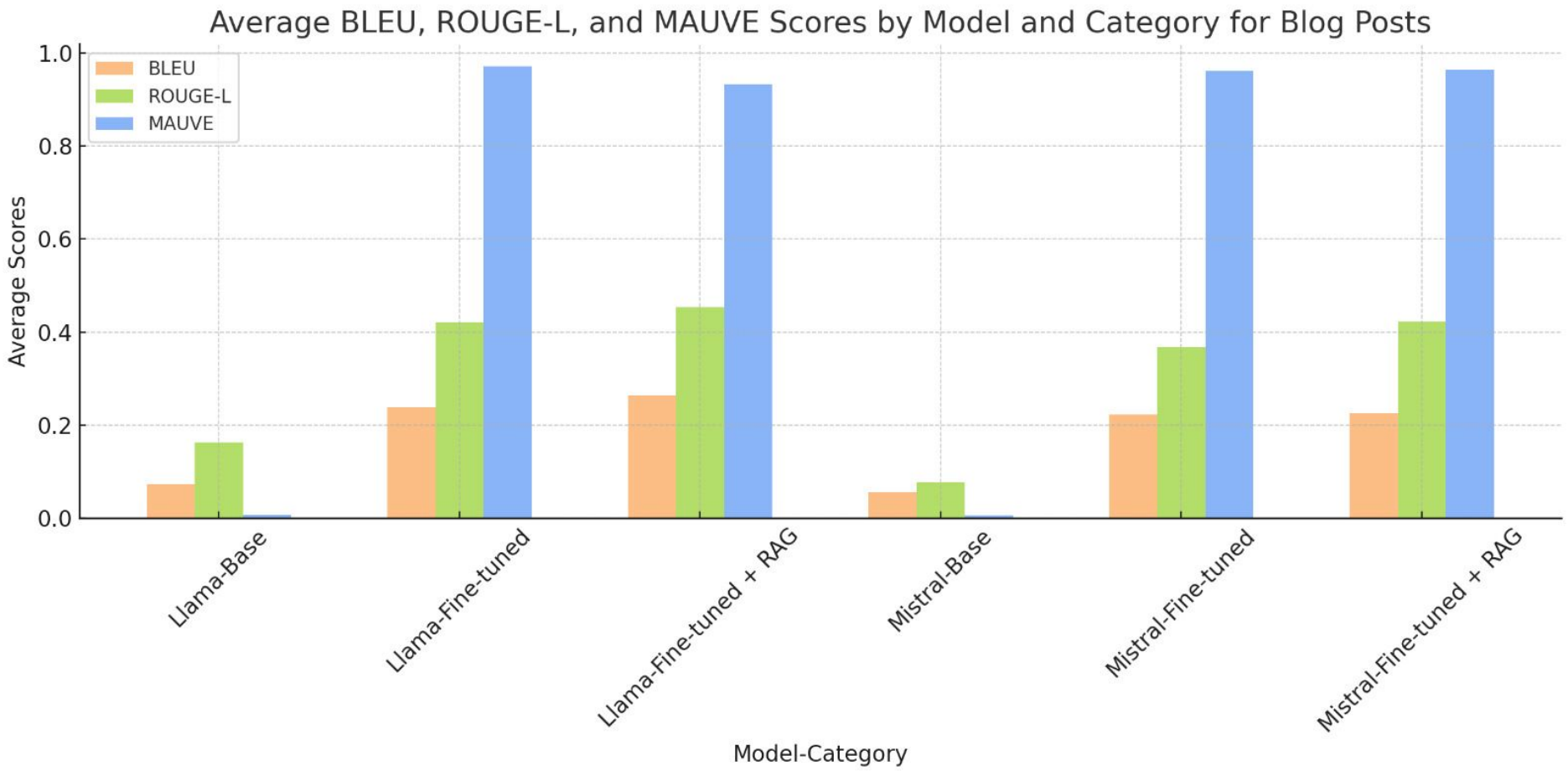
MAUVE Score:

- It quantifies the gap between the generated and human-written text distributions.
- It can capture distributional features such as sentence length, specific word frequencies, syntactic structures and semantic patterns.
- It also accounts for variability in the generated text, which is important for creative writing styles.

Results - Blog Posts

USER 1	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	0.0634	0.2253	0.2492	0.0843	0.2652	**0.2762**
ROUGE-L	0.1421	0.4083	0.3972	0.1734	0.4274	**0.4463**
MAUVE	0.0073	0.9768	0.9744	0.0094	**0.9842**	0.9274

USER 2	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	0.0482	0.2187	0.2017	0.0628	0.2113	**0.2503**
ROUGE-L	0.0127	0.3281	0.4482	0.1523	0.4133	**0.4618**
MAUVE	0.0045	0.9462	0.9523	0.0056	**0.9584**	0.9361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MAUVE
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0.4
0.2
0.0
Llama-Base
Llama-Fine-tuned
Llama-Fine-tuned + RAG
Mistral-Base
Mistral-Fine-tuned
Mistral-Fine-tuned + RAG
Model-Category

Example - Blog Posts

Reference:
Well boy did I have a bad day. Although you could say bad afternoon really. The whole us lot getting banned from the tech block has really peed me off, I mean, we don't exactly trash the place!!!!!!! Grrrrr. Where are we suposed to go now though huh? Our old corner? Thats a bit cold and wet. Ew. Fair enough if it were sunny but heloooooo this is england! We don't do sun! Much. Then bloody Chris in english flicking that candy off those candy necklasses thing. He was trying to hit Lee but got me instead an dit bloody well stung like mad. If I had looked up just at that moment it would've got me in the eye. Then when I anwsered Sir's questions and it took like ten minutes just for people to write it down. Geez. Now I am cream-crackered and wondering where our Jo has gone. Jo? Where you ma Freddie friend? And Jackie, I got ma Here On Earth DVD! Yay.

Base model:
It's safe to say that today has been a challenging day filled with unexpected obstacles and frustrating moments. As I sat down at my usual spot in the tech block, my screen died, leaving me stranded and cut off from the world of technology. As if that wasn't bad enough, our communication department decided to move our weekly meeting to a new location, making it yet another hassle to add to my overflowing plate.

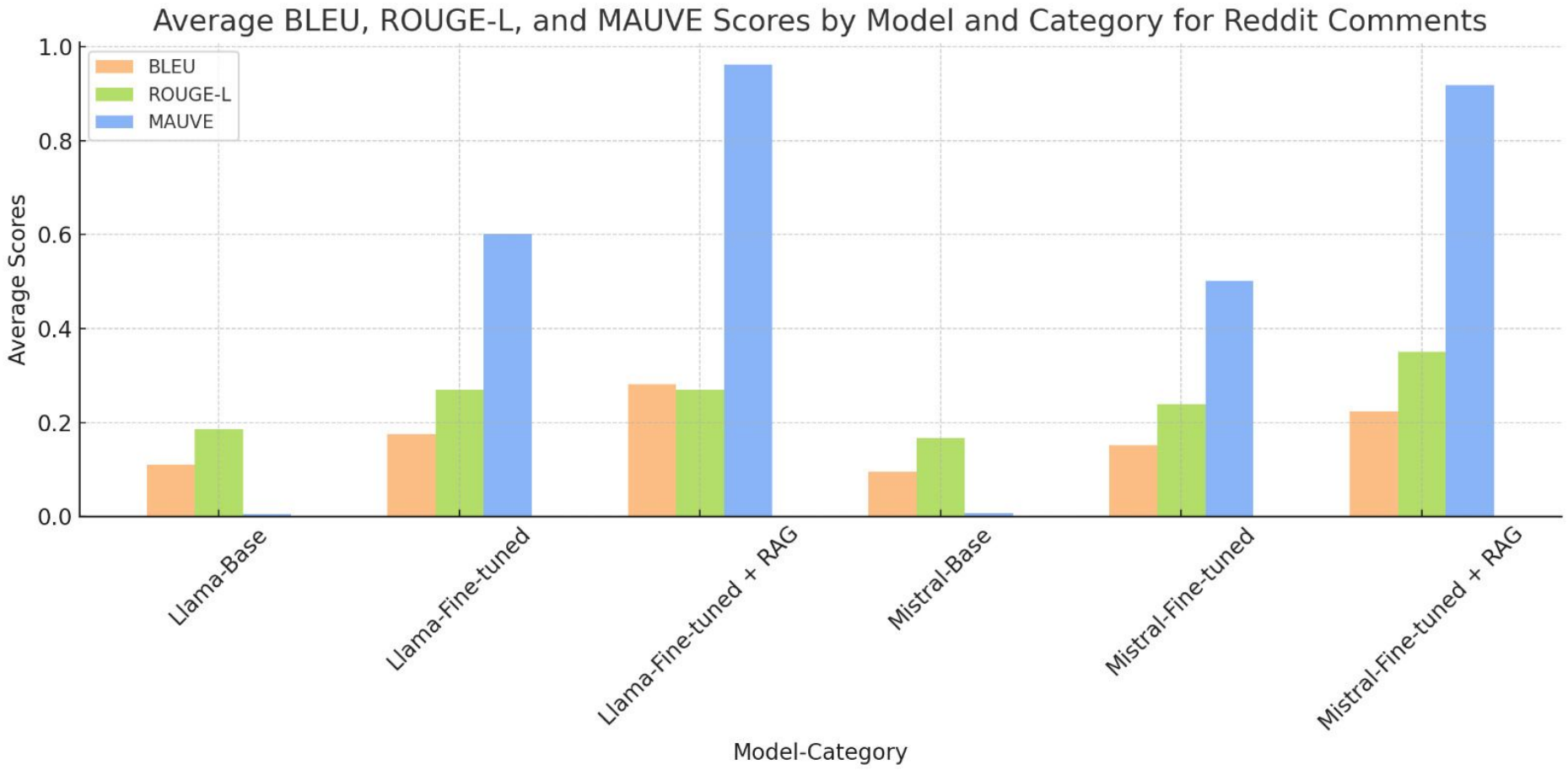
FT model:
I had my first lesson in there today and got told off. For what? Being bored! Oh well, i didn't do anything wrong anyway so it doesn't matter too much. The good thing is that we are moving our meetings to Room F12 next week because Room C3 has been taken over by the Art department (which is probably for the best). Yesterday was just one big disaster. I went out with Tom during lunchtime, but he fell over whilst trying to get into his car and ended up cutting his head open on the sharp edge of the door handle. So he came back to school bleeding everywhere. He couldn't go to his afternoon lessons either which meant he missed all his work and I wasn't able to see him. Then I stayed behind after my last lesson to help out and answer some questions, only to find myself locked inside the room until someone comes along to let me out. It was quite cold as well so I was freezing my little toes off. I am not impressed at all!!!!!

FT + RAG model:
lol nicki your doing fine in art, it was very funny when the dame messed up, and also when he started laughing, thats a good sign, it means he likes you :D the panto party sounds good fun, cant wait, i might be able to make it as well, ill see how i get on with the work i have today and then ill let you know. Well i have just come home from a very long and frustrating day. First of all, i had to leave the technology block because of my detention yesterday, so that was very annoying. Then, we had a geography lesson in there and our teacher forgot to book it, so we had to move. After that, i had a german lesson in there, and guess who turns up late? The person we were supposed to replace, so we had to wait for them to turn up. Very annoying!!!!!

Results - Reddit Comments

USER 1	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	0.0934	0.1571	0.2773	0.1081	0.1675	**0.3822**
ROUGE-L	0.1648	0.2475	0.4425	0.1828	0.2704	**0.2715**
MAUVE	0.0057	0.5602	0.9449	0.0045	0.2241	**0.9563**

USER 2	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	0.0966	0.1453	0.1695	0.1132	0.1845	**0.1813**
ROUGE-L	0.1681	0.2292	0.2579	0.1865	**0.2696**	0.2675
MAUVE	0.0085	0.4427	0.8918	0.0051	0.9794	**0.9689**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Model-Category

Example - Reddit Comments

Reference:
i use the atlanta public transit system . . . so pretty sketchy at times . especially when you ' re a skinny little white chick traveling alone . my parents don ' t know i use it . i want to practice psychology . specifically , i want to council veterans with ptsd . i wanted to join the military most of my childhood , and because of medical issues could not , so working with the military is the next best thing in my book . my parents would not disown me . my extended family probably would . it took years

Base model:
I've been using a Hypersteep flash wand for a while now and I am in love with it! Compared to other wands, it's much more lightweight and easy to handle. The fine-tipped head allows me to make the perfect microfoam for my latte art, and the ease of cleaning is a game-changer.

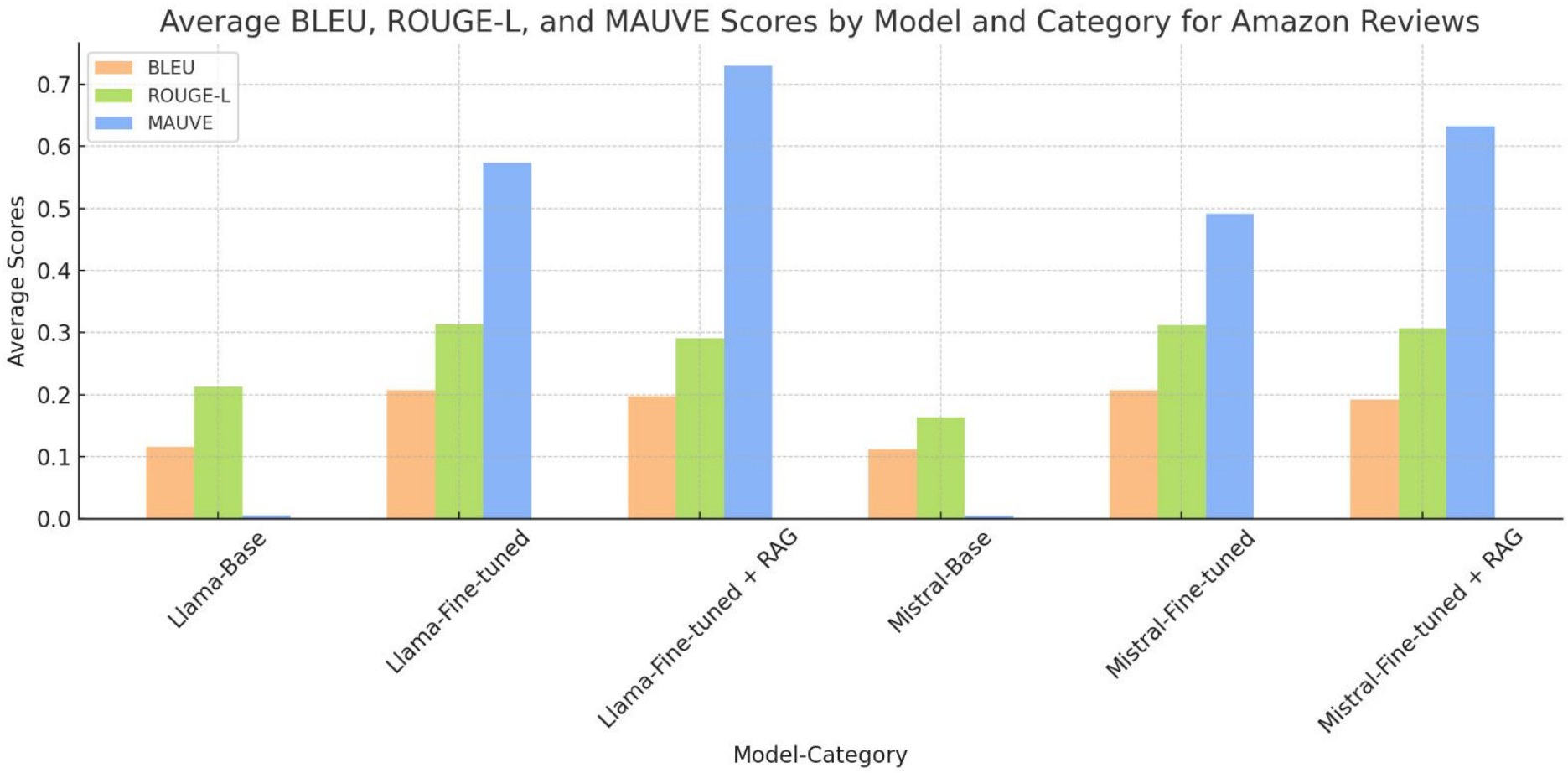
FT model:
my father was in the military and my grandfather was as well , but i have no personal experience with it . i have never been in combat or had to deal with anything like that . i can ' t speak for all military families , but from what i have seen , they are incredibly supportive of each other and their service members . they have to be to get through those long deployments . i believe that the military is an important institution that provides necessary protection for our country . i also believe that it needs to be reformed in order to better protect our troops.

FT + RAG model:
i actually want to go back into military service if they would have me . i don ' t think i could handle combat , but i ' d like to help out in some capacity . i ' d also like to see the world through their eyes . i grew up around it and have always been fascinated by it , even though i couldn ' t do it myself . i ' ve worked with veterans and it breaks my heart to see what they ' re going through . i ' d like to make a difference in that area if i can . i ' m also in school to become a therapist , so maybe that ' s where i ' ll end up .

Results - Amazon Reviews

USER 1	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	0.1279	0.2388	0.2362	0.1245	**0.2476**	0.2338
ROUGE-L	0.1870	0.3563	**0.3836**	0.2427	0.3513	0.3271
MAUVE	0.0054	0.4212	0.7202	0.0048	0.9218	**0.9790**

USER 2	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	0.0962	**0.1752**	0.1473	0.1082	0.1671	0.1599
ROUGE-L	0.1398	0.2674	0.2306	0.1827	**0.2751**	0.2534
MAUVE	0.0040	**0.5618**	0.5452	0.0055	0.2243	0.4816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0.4
0.5
0.6
0.7
Llama-Base
Llama-Fine-tuned
Llama-Fine-tuned + RAG
Mistral-Base
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Model-Category

Example - Amazon Reviews

Reference:
You can order more from Amazon or the Nespresso website or app. The sample pods included are a great selection of size and flavor options. Know that you cannot use the Originalline espresso pods in this - they are more of a cylinder shape, not the domes that this takes. Use the Virtuoline pods only with this. You seriously will not go back to regular coffee, especially Keurig, again! This stuff is thick and rich, and just unmatched anywhere. If you don't like the crema layer (foamy stuff) on your coffee or espresso, you might not like this, but if you stir it in, it's not as noticeable. If you love a good crema layer, you just can't beat a Virtuo machine.

Base model:
The Nespresso VirtuoPlus Deluxe Coffee and Espresso Machine offers a seamless and sophisticated coffee experience with its intuitive one-touch operation. It caters to a wide array of preferences, thanks to its extensive collection of compatible Nespresso Grand Cru pods. However, it's worth noting that this model may not be compatible with standard Nespresso pods, which could potentially limit your options if you have existing pods.

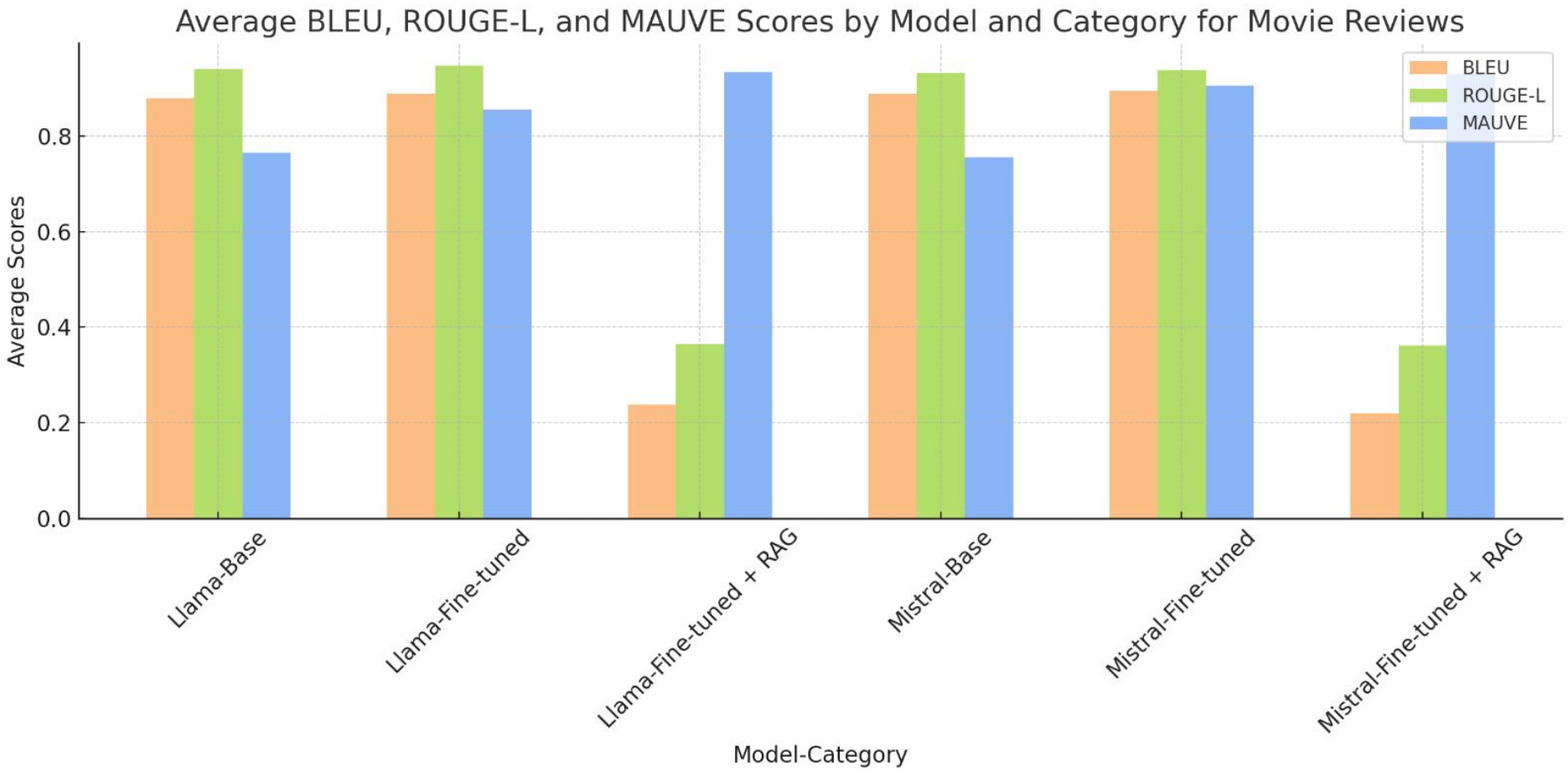
FT model:
I love the simplicity and user-friendly design of the Nespresso VertuoPlus Deluxe Coffee and Espresso Machine. It’s got a wide range of flavors, but note that it may have compatibility issues with some non-Nespresso pods. The quality of coffee and espresso is exceptional! It ensures the perfect coffee extraction every time, resulting in a rich and creamy crema. Aside from the ease of use and exceptional quality, the VertuoPlus Deluxe is sleek and stylish, making it a great addition to any modern kitchen.

FT + RAG model:
Buy the Nespresso VertuoPlus Deluxe Coffee and Espresso Machine for exceptional coffee experiences featuring a variety of delicious flavors, including unique blends such as Ristretto and Espresso, and enjoy a generous crema layer. Note that this model requires specific VirtuoLine pods, which may not be widely available in stores; however, purchasing directly from Nespresso ensures access to high-quality capsules. While occasional issues may arise due to product design flaws causing misaligned pods, these instances are rare.

Results - Movie Reviews

USER 1	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	0.8815	**0.8887**	0.1934	0.8625	**0.8776**	0.2376
ROUGE-L	0.9291	**0.9330**	0.3250	0.9353	**0.9485**	0.3295
MAUVE	0.7508	0.8953	**0.9241**	0.7689	0.7832	**0.9287**

USER 2	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	0.8953	**0.8992**	0.2456	0.8951	**0.9011**	0.2390
ROUGE-L	0.9361	**0.9420**	0.3985	0.9457	**0.9467**	0.3985
MAUVE	0.7601	0.9168	**0.9333**	0.7623	0.9285	**0.9381**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Model-Category

Example - Movie Reviews

Base model: Write a paragraph review for the movie 'Inception' in the style of a movie critic.
Inception, directed by Christopher Nolan, is a mind-bending science fiction thriller that will leave audiences questioning the nature of reality. The film follows the story of Dominic Cobb, played by Leonardo DiCaprio, a skilled thief who specializes in extracting valuable secrets from people's dreams. When Cobb is offered the chance to erase his criminal record in exchange for performing an unprecedented task, he must assemble a team of experts and delve deep into the subconscious of his target to plant an idea instead of stealing one. The visuals in this film are stunning, and the plot is filled with twists and turns that keep the audience on the edge of their seats. The performances are top-notch, with Ellen Page, Joseph Gordon-Levitt, and Tom Hardy delivering standout performances. Overall, Inception is a must-see for fans of science fiction and thrillers, and a testament to Nolan's talent as a director.

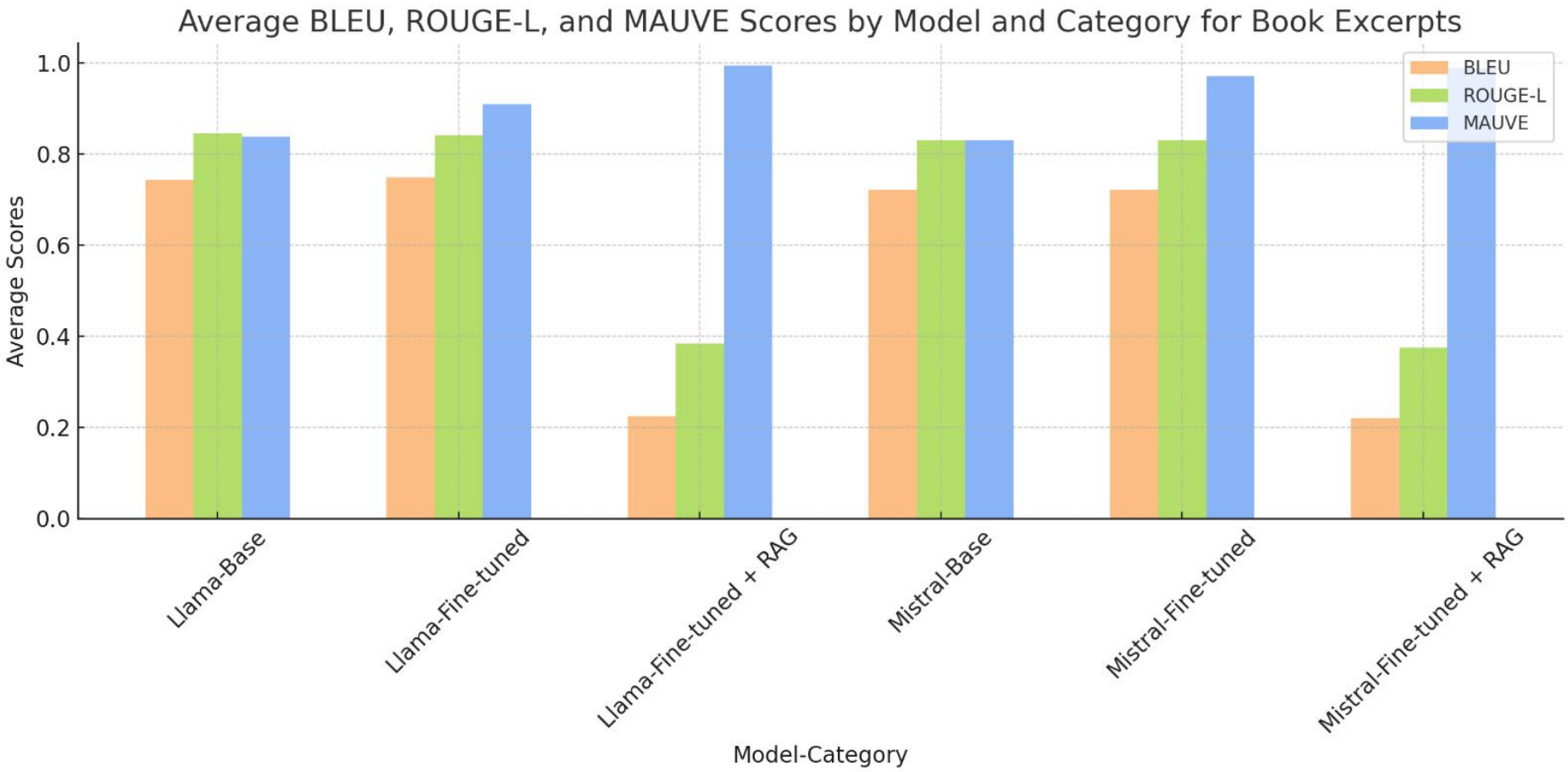
FT model: Write a paragraph review for the movie 'Inception' in the style of a movie critic.
Inception, directed by Christopher Nolan, is a mind-bending science fiction thriller that will leave audiences questioning the nature of reality. The film follows the story of Dominic Cobb, played by Leonardo DiCaprio, a skilled thief who specializes in extracting valuable secrets from people's dreams. When Cobb is offered the chance to erase his criminal record in exchange for performing an unprecedented task, he must assemble a team of experts to infiltrate the subconscious of his target and plant an idea in his mind. The visuals in this film are stunning, and the plot is filled with twists and turns that keep the audience on the edge of their seats. The performances by the cast are top-notch, with **DiCapario** delivering a standout performance as the troubled Cobb. Overall, Inception is a must-see for fans of science fiction and thrillers, and a testament to Nolan' talent as a director.

FT + RAG model: Write a paragraph review for the movie 'Inception' in the style of a movie critic using a critical tone.
Inception is a visually stunning and ambitious film that tries to explore complex ideas in a unique way. However, the execution of the concept falls short of expectations, and the script is poorly written. The cast is weak, and the characters lack depth, making it difficult to connect with them on an emotional level. The film also suffers from pacing issues, with some scenes feeling slow and unengaging. Overall, while the visual effects are impressive, the lack of a strong narrative and cast leaves you with a feeling of wanting something more.

Results - Book Excerpts

BOOK 1 - David Copperfield	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	**0.7227**	**0.7227**	0.2286	0.7384	**0.7394**	0.2384
ROUGE-L	**0.8392**	**0.8392**	0.3736	0.8389	**0.8458**	0.3857
MAUVE	0.8317	0.9749	**0.9938**	0.8451	0.8521	**0.9938**

BOOK 2 - Little Women	Mistral base	Mistral Fine-tuned	Mistral Fine-tuned + RAG	Llama base	Llama Fine-tuned	Llama Fine-tuned + RAG
BLEU	**0.7211**	**0.7211**	0.2120	0.7489	**0.7588**	0.2107
ROUGE-L	0.8231	**0.8238**	0.3782	0.8523	**0.8357**	0.3815
MAUVE	0.8289	0.9684	**0.9846**	0.8305	0.9669	**0.9957**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Mistral-Base
Mistral-Fine-tuned
Mistral-Fine-tuned + RAG
Model-Category

Example - Book Excerpts

Base model: Write a paragraph introduction for a book about a dog in the style of Jack London.
In the vast, untamed wilderness of the Yukon, where the wind howls like a pack of wolves and the snow blankets the earth in a shroud of white, there lived a dog. A dog of indomitable spirit and unyielding loyalty, whose very existence was a testament to the resilience of life in the face of adversity. This is the story of that dog, a tale of survival and triumph, of love and loss, and of the indelible bond between man and beast. It is a story that will touch the hearts of all who read it, and remind us of the power of the spirit that dwells within us all.

FT model: Write a paragraph introduction for a book about a dog in the style of Jack London.
In the great wilderness, where the law of nature holds sway over all living things, and man is but a transient visitor, there dwelt a canine soul of such depth and strength that it left its mark upon the hearts of those who encountered it. This is the story of that dog - a tale of endurance and courage, of love and loyalty, of a spirit unbroken by the cruel hand of fate. In the bleak and forbidding landscape of the wild, where even the trees seemed to keep their distance from one another, fearing the loneliness that lay in wait for them, he journeyed on, undaunted by the vastness of the world that stretched out before him. He knew not where his path would lead him, nor what trials and tribulations he would have to face along the way; all that was written in the stars above, and in the heart that beat within his sturdy chest.

FT + RAG model: Write a paragraph introduction for a book about a dog in the style of Jack London using a mysterious tone.
In the dimly lit room there slept the great animal; and outside the room, in the gloom of the basement, there was a strange impression that the animal slept. It was an impression that seemed to originate in a special phase of the mind - the mind in the act of sleeping. Few things can be more unutterable than the strange impression that the animal sleeps; and few things can be more inexplicable than the awakening of it. The sagacious reader may, if he will, conjecture what this animal is; but he cannot conjecture what it is to wake. When the animal wakes, something besides the animal wakes - something that has no name, **and never ithers,** "What is it that wakes?" whispers in the darkness.

Discussion

- Base models show low BLEU, ROUGE-L, and MAUVE scores.
- Fine-tuning significantly improves scores for both Llama and Mistral models, showing better alignment with reference data, domain-specific adaptation and diversity of generated text.
- BLEU and ROUGE using the same query and reference favors copying behavior, while MAUVE favors semantic diversity.Could explain why for FT + RAG, BLEU and ROUGE scores decreased in some cases while MAUVE increased.
- Improvement is seen across all datasets we used.
- Including RAG improves context relevance significantly in most cases, with the exception of a few.
- Mistral and Llama seem to offer similar quality of results.
- Used Google Colab credits for access to a more powerful GPU.

Challenges

- Requires a large number of high-quality writing samples
- High computational costs for user-specific model fine-tuning
- Evaluating stylistic alignment is subjective
- Ethical concerns: misuse of personalized models
- Stylistic variability across domains
- Lack of real-time feedback for dynamically adaptability
- Scaling the system is challenging

Future Scope

- Use more evaluation metrics that assess semantic similarity such as BERTScore, BLEURT.
- Fine-tune with data from human evaluators that score outputs based on relevance, coherence, and fluency rather than surface similarity to reduce copying.
- Train a single model per user across various domains that can adapt its response based on the domain.
- Incorporate additional adapter layers to add specific styles to the responses such as humor, verbosity or sarcasm.

Thank You!